

Maxwell DeFanti

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Applied Mathematics + Data Science

Berkeley, CA

Aug. 2024 – May 2028

- Probability and Random Processes, Probability Theory, Data Structures and Algorithms, Differential Equations, Abstract Algebra, Linear Algebra, Discrete Mathematics, Calculus 1-3, Data Science Foundations, Quantum Information Theory, Statistical Measure Theory and Real Analysis, Chemistry 1, Kinematics,

EXPERIENCE

Deep Learning Researcher

Berkeley Artificial Intelligence Research

September 2025 – Present

Berkeley, CA

- Designed and built novel Continual Compress Refresh algorithm with **PyTorch** that **completely eliminates loss of plasticity** in standard Feed-Forward, Convolutional, and Residual **neural networks** while outperforming the state of the art in task recall
- Developed a **deep learning compression algorithm** to prune redundant activation signals in **neural networks** into single nodes with a projected increase in inference efficiency of over **1000%** for high dimensional neural networks

Machine Learning Engineering Intern

Capital One

May 2026 – August 2026

Richmond, VA

- **Developing and training** a machine learning model to eliminate the need for manual queries of the Capital One log ingestion platform
- Working on the **Cloud Operations Resilience Engineering (CORE)** team to develop standardized, self-service query frameworks for **AWS S3**

Software Engineering Intern

The Materials Project

June 2025 – May 2026

Lawrence Berkeley National Lab, Berkeley, CA

- Built critical **full-stack applications** for admin side **database management**, filtration, and modification using **dash.py** and the **MPContribs** api for a DOE funded research initiative with **30,000+ citations**
- Developed a novel web interface that enabled **600,000+** users of the **Materials Project** to upload and modify their research data, implementing custom **React** components, efficient information extraction across file types and flexible data formatting with **Pandas**, and seamless **API** integration

Chief Machine Learning Engineer

GovernAIce

June 2025 – September 2025

Hybrid

- Led the development of a **RAG** model capable of synthesizing complex AI regulation compliance requirements. Used **LangChain** for text chunking, built a multilingual textual **vector embedding** model adapted to legal documents. Engineered data pipelines in **MongoDB**, **Finetuned LLMs** for legal accuracy and relevancy

PROJECTS

Activation Probes for Synthetic Toxin Variant Detection | *Python, PyTorch, ESM-2, Scikit-Learn*

April 2026

- Trained a **two-layer MLP activation probe** on frozen **ESM-2 650M** mean-pooled embeddings of natural and synthetic toxins using weighted binary cross-entropy, **AdamW**, and **CUDA**-accelerated training and inference
- Achieved **86.7% recall** on synthetic toxin variants at ~40% sequence identity where BLAST collapsed to **46.7%**, with an overall **AUC-ROC** of **0.983**

Multilevel LLM Jailbreak Security | *Python, PyTorch, Huggingface Transformers*

May 2026

- Implemented **Greedy Coordinate Gradient** suffixes as well as other common jailbreak attack strategies in a testing environment against a **2.5 billion parameter Qwen model AdamW**, and **CUDA**-accelerated training and inference
- Achieved **87.0% improvement** in jailbreak refusal rate over the base model identity with a **custom multilevel defense algorithm** that takes advantage of **self reminding**, **perplexity filtration**, and **LLM judging**

TECHNICAL SKILLS

Areas: Computer Vision, Data Science, Full-Stack Development, LLM Finetuning, Financial Markets, Machine Learning, Mathematical Modeling, Natural Language Processing, Optics, Software Engineering, Statistics, Deep Learning, CNN, ResNet

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, MATLAB, SQL, Linux

Frameworks: React, Node.js, Flask, Plotly Dash

Developer Tools: Git, VS Code IDE, Visual Studio, Jupyter Notebook

Libraries: PyTorch, Scikit-learn, Cuda, Pandas, NumPy, Huggingface Transformers, Matplotlib, OpenCV, MATLAB Engine, NLTK, LangChain, MPContribs, PyMOL